IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

SIGNALS AND SYSTEMS

Wednesday 10 May 14:30

Time allowed: 2 hours

There are THREE questions on this paper.

Answer ALL questions.

Any special instructions for invigilators and information for candidates are on page 1.

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Signals and Systems

Formulae

Standard Laplace Transforms

Signal	Transform	ROC
$u(t)$	$\frac{1}{s}$	$\mathrm{Re}(s) > 0$
$tu(t)$	$\frac{1}{s^2}$	$\mathrm{Re}(s) > 0$
$\delta(t-\tau),\ \tau \geq 0$	$e^{-s\tau}$	$\forall s$
$e^{-at}u(t)$	$\frac{1}{s+a}$	$\mathrm{Re}(s) > -a$
$te^{-at}u(t)$	$\frac{1}{(s+a)^2}$	$\mathrm{Re}(s) > -a$
$\cos(w_1 t)u(t)$	$\frac{s}{s^2+w_1^2}$	$\mathrm{Re}(s) > 0$
$\sin(w_1 t)u(t)$	$\frac{w_1}{s^2+w_1^2}$	$\mathrm{Re}(s) > 0$
$e^{-at}\cos(w_1 t)u(t)$	$\frac{s+a}{(s+a)^2+w_1^2}$	$\mathrm{Re}(s) > -a$
$e^{-at}\sin(w_1 t)u(t)$	$\frac{w_1}{(s+a)^2+w_1^2}$	$\mathrm{Re}(s) > -a$

1. *This question caries 40% of the marks.*

a) For each of the following continuous-time signals, state whether they are causal or noncausal, periodic or nonperiodic, and even, odd or non-symmetric. Give justifications for your statements. If periodic, state the period.

i) $x(t) = 1 + |\cos(\pi t)|$ [2]

ii) $x(t) = 2\sin^2(2t - \pi/2)$ [2]

iii) $x(t) = \begin{cases} e^{-2t/3} & t \geq -1 \\ 0 & \text{otherwise.} \end{cases}$ [2]

b) Consider the signal

$$x(t) = \begin{cases} t-1 & 0 \leq t \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

For each of the following signals $y(t)$, find an expression for $y(t)$ not involving x. Describe the sequence of steps that must be taken to form $y(t)$ from $x(t)$. Show the sequence of steps in illustrative sketches.

i) $y(t) = x(t + 3/2)$ [3]

ii) $y(t) = x(-2t + 1)$ [3]

c) An LTI system has input signal $x(t)$ and output signal $y(t)$, with Laplace transforms $X(s)$ and $Y(s)$ respectively. The system has the transfer function

$$\frac{Y(s)}{X(s)} = \frac{3}{s+3}.$$

i) Draw a labelled sketch describing any poles and zeros of this system. [2]

ii) Write down an expression relating $y(t)$ and $x(t)$. [2]

iii) Find the impulse response of this system, assuming it is stable. [2]

d) Consider an LTI system with system function $H(s)$. Assuming the Region of Convergence of $H(s)$ exists, state whether the Region of Convergence of $H(s)$ is (choose any that are true): [3]

A. a region with edges parallel to the vertical axis

B. a region with edges joining the points at different frequencies using spline interpolation

C. a region with edges forming a circle around the origin

D. a region not containing any roots of the system function

E. a region never containing the origin

e) i) Write down a general expression in the z-transform domain for a Finite Impulse Response filter of order M. [3]

ii) Design a Finite Impulse Response notch filter of order 2 for which the notch is at a frequency of 2.5 kHz. The filter operates at a sampling frequency of 10 kHz. [4]

f) Consider the signal $x_1[n]$ shown in Fig. 1.1. The magnitude spectrum, $|X_1[k]|$, is computed using a 12-point DFT and is shown in Fig. 1.2. The sampling frequency used in this question is 48 kHz.

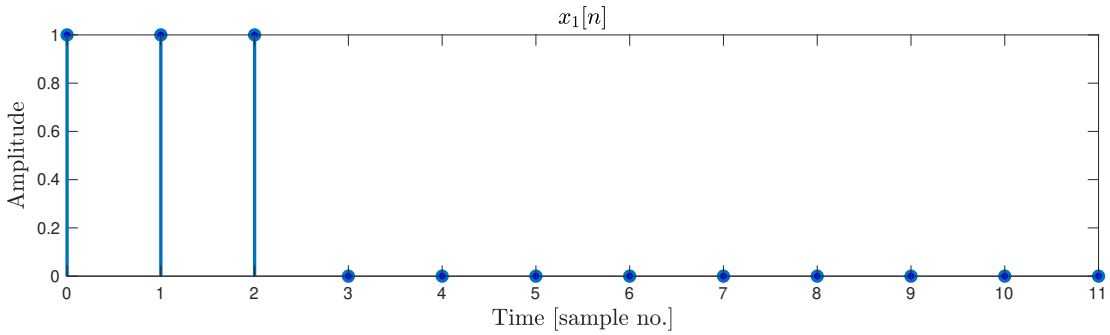


Figure 1.1

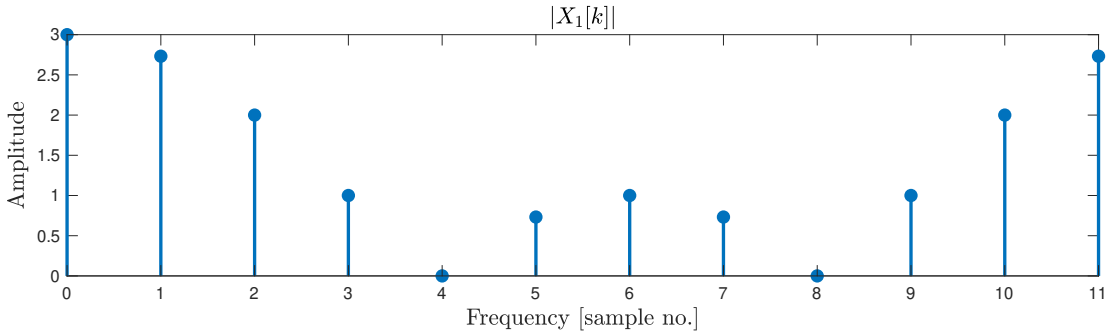


Figure 1.2

i) What is the frequency resolution, Δf of the 12-point DFT? [1]

ii) Is the frequency domain representation of the signal $X_1[k]$ purely real, purely imaginary, or complex? What can you conclude about the phase of $X_1[k]$? [3]

iii) Now consider the signal $x_2[n]$ plotted in Fig. 1.3. The 12-point DFT is taken to obtain $X_2[k]$. Describe similarities and differences between $|X_2[k]|$ and $|X_1[k]|$, stating any property of the Fourier Transform supporting your argument. [4]

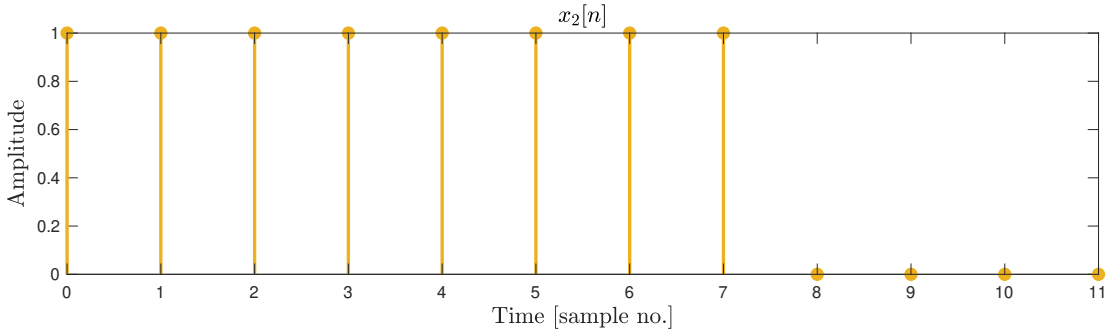


Figure 1.3

iv) You now would like to remove high-frequency components in $x_1[n]$. Someone suggested setting $X_1[k] = 0$ for $k = k_c \ldots 11$, where k_c is the index of the desired cut-off frequency, and then taking the inverse DFT to obtain $x_{lp}[n]$. What is wrong with this approach? Suggest a more appropriate method. [4]

2. *This question caries 30% of the marks.*

a) Consider the classes of continuous-time signals and discrete-time signals. For each, further consider the classes of periodic signals and nonperiodic signals.

State the Fourier representation appropriate for each class of signal and give the corresponding formulae to obtain the Fourier representation from the time domain signal. [4]

b) One period of a particular periodic signal with period $T = 3$ s is defined as

$$x(t) = 1 - e^{-2t} \quad 0 \le t \le T.$$

i) Sketch $x(t)$ for $0 \le t \le 8$ s. [3]

ii) Derive an appropriate Fourier representation of $x(t)$. [3]

iii) Hence determine the magnitude spectrum of the signal at $1/3$ Hz. [3]

c) The frequency response of a particular continuous-time system is

$$H(j\omega) = \frac{10}{j\omega + 10}$$

and the input signal is $x(t) = \cos(4\pi t)$.

i) Write down the Fourier representation of the input signal.[3]
Hint: consider using terms of the form $\delta(\omega + \ldots)$.

ii) When $x(t)$ is used as the input signal, find the Fourier representation of the output signal, and write it in the form $A(R + jI)$ where A is a scalar, and R and I are the real and imaginary terms respectively. [3]

iii) Hence find the time domain output signal, $y(t)$. [2]

d) Consider a continuous-time LTI system characterized by the system function $H(s)$. When the input signal to the system is $x(t) = \delta(t)$, the corresponding output signal is

$$y(t) = \frac{2}{3}e^{3t}u(-t) + \frac{2}{3}e^{-t}u(t).$$

i) Find $H(s)$ and state the Region of Convergence. [4]

ii) Sketch a representation of $H(s)$ on the s-plane and indicate the Region of Convergence. [2]

iii) The input signal is now changed to the unit step function so that $x(t) = u(t)$. Find the Laplace transform of the corresponding output signal and state its Region of Convergence. [3]

3. *This question caries 30% of the marks.*

a) The discrete-time signal $x[n]$ is defined as

$$x[n] = \begin{cases} 1 & 0 \leq n < 5 \\ 0 & \text{otherwise.} \end{cases}$$

i) Find the DTFT of $x[n]$ and express the DTFT of $x[n]$ as a ratio of sine functions. [8]

ii) Sketch the magnitude and phase of the DTFT of $x[n]$. [3]

b) The sound of a large bell known as Big Ben is to be recorded using a microphone and a digital sound recording device with variable input gain.

i) Describe how you would determine the minimum sampling frequency fs Samples/s that should be used for the recording, including a brief description of any measurements or parameters that would be needed to detemine fs. [3]

ii) Assume a recording has been made at an appropriate sampling frequency fs Samples/s, and a histogram of the samples has been plotted as represented in Fig. 3.1.

Explain the peak in the histogram at zero amplitude. [1]

Explain the peaks in the histogram at amplitudes of ± 1000. State whether these peaks are an expected characteristic of this recording and suggest a reason they occur in the histogram. [4]

Explain what could be done so that the peaks at amplitudes of ± 1000 would not occur in the histogram. [2]

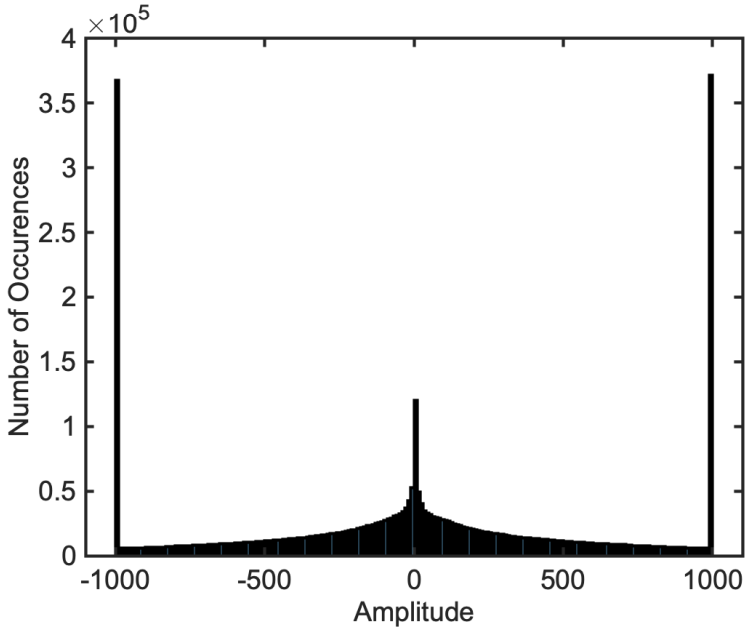


Figure 3.1

c) Consider a discrete-time LTI system with impulse response $h[n]$ and

system function $H(z)$ with

$$h[n] = [1,\ -0.1,\ 0.12,\ -0.04].$$

i) Write an expression for $H(z)$. [1]

ii) Find the magnitudes of $H(z)$ at normalized angular frequencies of 0 and π, and express these in decibels. [3]

d) Consider an LTI system described in the z-transform domain by the system function

$$H(z) = \frac{3 - \frac{5}{6}z^{-1}}{(1 - \frac{1}{4}z^{-1})(1 - \frac{1}{3}z^{-1})} \quad \text{for } |z| > \frac{1}{3}.$$

Find the impulse response of the system, $h[n]$. [5]
Hint: consider using partial fraction expansion